

National Exam of Higher National Diploma 2022 Session

Spécialty/option: Software Engineering (SWE)
Paper: Information System
Duration : 3 Hours

Credits : 6

SECTION A: SYSTEM ARCHITECTURE 40 MARKS

1. What are the essential characteristics of cloud computing? (4mks)
2. How many types of deployment models cloud? (4mks)
3. Describe about the different Guided Medias. (4mks)
4. What do you mean by wireless communication? (4mks)
5. List two (02) advantages and two (02) disadvantages of fiber optics cable. (4mks)
6. List three (03) factors on which data rate depends. (4mks)
7. What is an Object server in client server environment? (4mks)
8. What are the two broad classes of middleware in client server environment? (4mks)
9. What are the five major technologies that can be used to create Client/Server applications in client server environment? (4mks)
10. What is *public cloud* and *private cloud*? (4mks)

SECTION B: OPERATING SYSTEMS (40 MARKS)

1. Why are user-level threads packages generally cooperatively scheduled? (3mks)
2. Describe a sequence the sequence of step that occur when a timer interrupt occurs that eventually results in a context switch to another application. (6mks)
3. Context switching between two threads of execution within the operating system is usually performed by a small assembly language function. In general terms, what does this small function do internally? (6mks)

4. What is a *race condition*? Give an example. (4mks)
5. What is a critical region? How do they relate to controlling access to shared resources? (5mks)
6. What are three requirements of any solution to the critical sections problem? Why are the requirements needed? (5mks)
7. Why is *turn passing* a poor solution to the critical sections problem? (4mks)
8. Interrupt disabling and enabling is a common approach to implementing mutual exclusion, what are its advantages and disadvantages? (4mks)

SECTION C: PROJECT MANAGEMENT AND LEGAL REGULATIONS (20 MARKS)

1. Define the following terms
 - a. Project (1 mark)
 - b. Software project manager (1mark)
2. Software project management comprises of a number of activities, which contains planning of project, deciding scope of software product, estimation of cost in various terms, scheduling of tasks and events, and resource management.
 - a. What are the activities that are carried out during project scope management activities (4marks)
 - b. Explain what is usually estimated during project estimation phase (8 marks)
 - c. Discuss two types of Project Estimation Techniques (2marks)
3. A. what is Project Scheduling (2marks)
B. what are the activities that are carried out during Project Scheduling (4marks)